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Inventor(s)

TEYSSEYRE; Jerome et al.

POWER MODULE PACKAGE FOR DIRECT COOLING MULTIPLE POWER MODULES

Abstract

According to an aspect, a power module package includes a plurality of power modules including a first power module and a second power module, a plurality of heat sinks including a first heat sink coupled to the first power module and a second heat sink coupled to the second power module, and a module carrier coupled to the plurality of power modules, where the module carrier includes a first region defining a first heat-sink slot and a second region defining a second heat-sink slot. The first heat sink extends at least partially through the first heat-sink slot and the second heat sink extends at least partially through the second heat-sink slot. The power module package includes a housing coupled to the module carrier and a ring member located between the module carrier and the housing.

Inventors: TEYSSEYRE; Jerome (Singapore, SG), YOO; Inpil (Unterhaching, DE), EOM; Jooyang (Gimpo-si, KR)

Applicant: SEMICONDUCTOR COMPONENTS INDUSTRIES, LLC (Scottsdale, AZ)

Family ID: 81588527

Assignee: SEMICONDUCTOR COMPONENTS INDUSTRIES, LLC (Scottsdale, AZ)

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/542,230, filed Dec. 15, 2023, which is a continuation application of U.S. application Ser. No. 17/822,844, filed Aug. 29, 2022, now U.S. Pat. No. 11,848,320, which is a divisional application of U.S. patent application Ser. No. 16/949,896, filed Nov. 19, 2020, all of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] This description relates to a power module package for direct cooling multiple power modules.

BACKGROUND

[0003] A power module package may contain multiple power modules. However, some conventional power module packages may be relatively large to meet cooling performance requirements.

SUMMARY

[0004] According to an aspect, a power module package for direct cooling multiple power modules includes a plurality of power modules including a first power module and a second power module, a plurality of heat sinks including a first heat sink coupled to the first power module and a second heat sink coupled to the second power module, and a module carrier coupled to the plurality of power modules, where the module carrier includes a first region defining a first heat-sink slot and a second region defining a second heat-sink slot. The first heat sink extends at least partially through the first heat-sink slot and the second heat sink extends at least partially through the second heat-sink slot. The power module package includes a housing coupled to the module carrier, where the module carrier is disposed between the housing and the plurality of power modules. The power module package includes a ring member located between the module carrier and the housing.

[0005] According to some aspects, the power module package may include one or more of the following features (or any combination thereof). The first heat sink includes metal pin fins. The first region may include at least one positioning guide configured to align at least one of the first power module or the first heat sink. The module carrier includes a plastic material. The plurality of power modules may include a third power module and the plurality of heat sinks may include a third heat sink coupled to the third power module. The module carrier includes a third region defining a third heat-sink slot, and the third heat sink extends at least partially through the third heat-sink slot. Each of the first power module, the second power module, and the third power module may correspond to a separate phase of a three phase current of an inverter. The ring member includes an O-ring that extends around the first power module, the second power module, and the third power module, collectively. The first power module includes a half-bridge module. The module carrier is coupled to the housing via one or more fasteners.

[0006] According to an aspect, a power module package for direct cooling multiple power modules

includes a plurality of power modules including a first power module, a second power module, and a third power module, a plurality of heat sinks including a first heat sink coupled to the first power module, a second heat sink coupled to the second power module, and a third heat sink coupled to the third power module, and a module carrier coupled to the plurality of power modules, the module carrier including a first region defining a first heat-sink slot, a second region defining a second heat-sink slot, and a third region defining a third heat-sink slot. The first heat sink extends at least partially through the first heat-sink slot. The second heat sink extends at least partially through the second heat-sink slot. The third heat sink extends at least partially through the third heat-sink slot. The power package module includes a housing coupled to the module carrier, where the housing is coupled to the module carrier being located between the housing and the plurality of power modules.

[0007] According to some aspects, the power module package may include one or more of the following features (or any combination thereof). The power module package may include a ring member located between the module carrier and a surface of the housing. The first region defines one or more module positioning guides. The first region defines one or more heat-sink positioning guides. Each of the first power module, the second power module, and the third power module corresponds to a separate phase of a three phase current of an inverter. Each of the first heat sink, the second heat sink, and the third heat sink include metal pin fins. The module carrier includes a metal-based material.

[0008] According to an aspect, a method of assembly a power module package integrating multiple power modules includes receiving a module carrier, the module carrier including a first region defining a first heat-sink slot and a second region defining a second heat-sink slot, applying an adhesive material to the first region and the second region, inserting a first heat sink coupled to a first power module through the first heat-sink slot until the first power module contacts the adhesive material, and inserting a second heat sink coupled to a second power module through the second heat-sink slot until the second power module contacts the adhesive material. In some examples, each of the first region and the second region defines one or more heat-sink positioning guides.

[0009] According to an aspect, a method of assembly a power module package integrating multiple power modules includes receiving a module carrier, the module carrier including a first region defining a first heat-sink slot and a second region defining a second heat-sink slot, inserting a first heat sink into the first heat-sink slot, inserting a second heat sink into the second heat-sink slot, applying an adhesive to the first region, the second region, the first heat sink, and the second heat sink, contacting a first power module to the adhesive on the first region and the first heat sink, and contacting a second power module to the adhesive on the second region and the second heat sink. Each of the first region and the second region defines one or more module positioning guides and one or more heat-sink positioning guides.

[0010] The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1A illustrates a cross sectional view of a power module package according to an aspect.

[0012] FIG. 1B illustrates a module carrier of the power module package according to an aspect.

[0013] FIG. 1C illustrates a top view perspective of the power module package according to an aspect.

[0014] FIGS. 2A through 2F illustrate a power module package according to an aspect.

[0015] FIGS. 3 and 4 illustrate examples of a heat sink according to several aspects.

[0016] FIG. 5A through 5F illustrate a power module package that is assembled according to an aspect.

[0017] FIG. 6 illustrates a flowchart depicting example operations for assembling a power module package according to an aspect.

[0018] FIGS. 7A through 7D illustrate a power module package that is assembled according to another aspect.

[0019] FIG. 8 illustrates a flowchart depicting example operations for assembling a power module package according to an aspect.

DETAILED DESCRIPTION

[0020] The present disclosure relates to a power module package for direct cooling multiple power modules. The power module packages discussed herein may be compact, robust, and/or relatively easy to assemble while providing an efficient structure to permit direct cooling. The power module package includes a plurality of power modules, a plurality of heat sinks coupled to the power modules, a module carrier coupled to the power modules, a housing coupled to the module carrier, and a ring member located (e.g., disposed) between the module carrier and the housing.

[0021] In some examples, the power modules include two power modules. In some examples, the power modules include three power modules. In some examples, the power modules include half-bridge power modules. The power module package may include a first power module, a second power module, and a third power module, where each module corresponds to a separate phase of a three-phase current (UVM) of an inverter. In some examples, the power modules include more than three power modules.

[0022] The power modules are integrated into the module carrier, and the module carrier is coupled to the housing such that the module carrier is disposed between the power modules and the module carrier. In some examples, the housing is a housing for an inverter. The ring member may be disposed within a groove on the housing, and may extend around the power modules, collectively. In some examples, the power module package includes a single ring member. The module carrier may define a number of regions, where each region defines a heat-sink slot. For example, the module carrier may include a first region defining a first heat-sink slot and a second region defining a second heat-sink slot. In some examples, each of the first region and the second region includes one or more positioning guides that properly align a power module and/or a heat sink on a respective region.

[0023] The power package module may be assembled according to a first assembly process.

According to the first assembly process, in a pre-processing step, the heat sinks are coupled to the power modules. For example, the first power module is coupled to a first heat sink, and the second power module is coupled to a second heat sink. In some examples, the heat sinks include pin fin heat sinks, where a pin fin heat sink includes a base plate with a plurality of pin fins that extend from the base plate. The base plate is coupled to a surface of a power module. According to the first assembly process, an adhesive material is dispensed on each region such as the first region and the second region. The first heat sink is inserted into the first heat-sink slot until the first power module contacts the adhesive material on the first region (where the positioning guide(s) are used to properly align the first power module on the first region), and the second heat sink is inserted into the second heat-sink slot until the second power module contacts the adhesive material on the second region (where the positioning guide(s) are used to properly align the second power module on the second region). Then, the module carrier with the power modules are cured (e.g., oven cured by heat).

[0024] The power package module may be assembled according to a second assembly process.

According to the second assembly process, the heat sinks are initially not coupled to the power modules, but rather the individual heat sinks are inserted into the heat-sink slots, and the power

modules are then coupled to the module carrier and the heat sinks. For example, The first heat sink is inserted into the first heat-sink slot (where the positioning guide(s) are used to properly align the first heat sink on the first region), and the second heat sink is inserted into the second heat-sink slot (where the positioning guide(s) are used to properly align the second heat sink on the second region. Then, an adhesive material is dispensed on the first region, the second region, the first heat sink, and the second heat sink. The first power module is positioned on the adhesive material on the first region and the first heat sink, and the second power module is positioned on the adhesive material on the second region and the second heat sink. Then, the module carrier with the power modules are cured (e.g., oven cured by heat).

[0025] FIGS. 1A through 1C illustrate a power module package **100** for direct cooling multiple power modules **102**. FIG. 1A illustrates a cross sectional view of the power module package **100**. The power module package **100** includes a plurality of power modules **102**, a plurality of heat sinks **110** coupled to the power modules **102**, a module carrier **104** coupled to the power modules **102**, a housing **106** coupled to the module carrier **104**, and a ring member **108** located (e.g., disposed) between the module carrier **104** and the housing **106**.

[0026] FIG. 1B illustrates an enlarged view of the module carrier **104**. The module carrier **104** is disposed between the power modules **102** and the housing **106**, where the module carrier **104** defines a number of heat-sink slots (e.g., **120-1**, **120-2**, **120-3**) that receive (and align) a separate heat sink **110**. FIG. 1C illustrates a top view perspective of the power module package **100** that depicts the ring member **108** extending around the power modules **102**, collectively. In some examples, the power module package **100** includes a single ring member **108**. The ring member **108** may be disposed between the module carrier **104** and the housing **106** and may assist with sealing a chamber **124** of the housing **106**.

[0027] The power module package **100** may be a high-voltage power package. In some examples, the power module package **100** may operate at voltages greater than one thousand volts. The amount of heat generated by the power modules **102** is reduced by injecting a cooling fluid into the chamber **124** of the housing **106**. The cooling fluid may include cooled gas such as air and/or cooled liquid such as a cooling fluid and/or cooling mixture. The power modules **102** are directly cooled by the cooling fluid. Direct cooling involves the application of a cooling fluid directly on a portion of a power module **102** (e.g., a backside of a power module **102** such as a heat sink **110** coupled (e.g., directly coupled) to the power module **102**). For example, since heat sinks **110** are coupled (e.g., directly coupled) to the power module **102**, the application of cooling fluid to the heat sinks **110** within the chamber **124** directly cools the power modules **102**. In contrast, indirect cooling involves the application of a cooling fluid to a component that is sealed from one or more power modules. The power module package **100** may be relatively compact, robust, and relatively simple to assemble. In some examples, instead of individually sealing each power module for direct cooling, the power module package **100** uses a module carrier **104** that can seal, collectively, multiple power modules **102** in a compact fashion.

[0028] A power module **102** includes one or more silicon powered dies soldered onto one or more substrates, which can be direct bonded copper (DBC) metalized ceramics soldered onto a base plate (e.g., a copper plate). In some examples, a power module **102** includes one or more silicon-based power chips (e.g., insulated-gate bipolar transistor(s) (IGBTs), metal oxide silicon field effect transistor(s) (MOSFETs), diodes, etc.) that are soldered onto a ceramic-based substrate, which is then soldered to a base plate. In some examples, the power modules **102** include half-bridge modules (e.g., an inverter leg with two switches and their corresponding antiparallel diodes). However, the power modules **102** may include other types of module topologies such as switch, bridge rectifier, H-bridge, etc.

[0029] In some examples, the power module package **100** includes half bridge direct cooling modules that are integrated on an inverter. In some examples, the power module package **100** includes a three-phase inverter circuit (e.g., a 6-pack module or IGBT 6-pack module). In some

examples, the power module package **100** is configured to be used within an automotive inverter for an electric vehicle or a hybrid vehicle. For example, an automotive inverter may accelerate or decelerate the vehicle by converting the direct current (DC) power from the batteries to alternating current (AC) at the frequency required for vehicle speed and other system controls to control the electric motor speed, drive, torque, and/or power. In some examples, the housing **106** is the housing of the automotive inverter.

[0030] The power modules **102** include a first power module **102-1** and a second power module **102-2**. In some examples, the first power module **102-1** is a half-bridge power module. In some examples, the second power module **102-2** is a half-bridge power module. In some examples, the power modules **102** include a third power module **102-3**. In some examples, the third power module **102-3** is a half-bridge power module. In some examples, the power module package **100** includes two power modules **102**. In some examples, the power module package **100** includes three power modules **102**. In some examples, the power module package **100** includes more than three power modules **102**. In some examples, each of the first power module **102-1**, the second power module **102-2**, and the third power module **102-3** correspond to a separate phase of a three-phase current (UVM) of an inverter. In some examples, the first power module **102-1**, the second power module **102-2**, and the third power module **102-3**, collectively, operate as a 6-pack full bridge module.

[0031] Each of the power modules **102** include a first surface **111** and a second surface **113**. The distance between the first surface **111** and the second surface **113** may define a thickness of a respective power module **102** in a direction **A1**. Each power module **102** is connected to a separate heat sink **110**. In some examples, each power module **102** is connected to a separate heat sink **110** using an adhesive material (e.g., a polymer-based material, solder material, glue material, etc.). The first power module **102-1** is coupled to a first heat sink **110-1**, the second power module **102-2** is coupled to a second heat sink **110-2**, and the third power module **102-3** is coupled to a third heat sink **110-3**. In further detail, the first heat sink **110-1** is coupled to the second surface **113** of the first power module **102-1**, the second heat sink **110-2** is coupled to the second surface **113** of the second power module **102-2**, and the third heat sink **110-3** is coupled to the second surface **113** of the third power module **102-3**.

[0032] The first heat sink **110-1** extends from the second surface **113** of the first power module **102-1** in the direction **A1**. The second heat sink **110-2** extends from the second surface **113** of the second power module **102-2** in the direction **A1**. The third heat sink **110-3** extends from the second surface **113** of the third power module **102-3** in the direction **A1**. In some examples, the first heat sink **110-1** includes a pin fin heat sink. In some examples, the second heat sink **110-2** includes a pin fin heat sink. In some examples, the third heat sink **110-3** includes a pin fin heat sink. In some examples, a pin fin heat sink includes a base plate that is coupled to the second surface **113** of a respective power module **102**, and a plurality of pin fins that extend in the direction **A1**.

[0033] Each power module **102** is coupled to the module carrier **104** via an adhesive material **112**. In some examples, the adhesive material **112** includes a polymer-based material. In some examples, the adhesive material **112** includes a solder material. In some examples, the adhesive material **112** includes a glue material. In some examples, the adhesive material **112** includes a transient liquid phase sintering (TLPS) material. The module carrier **104** includes a first surface **115** and a second surface **117**. In some examples, the second surface **117** is parallel with the first surface **115**. The distance between the first surface **115** and the second surface **117** may define the thickness of the module carrier **104** in the direction **A1**. The first surface **115** of the module carrier **104** is coupled to the second surface **113** of each of the first power module **102-1**, the second power module **102-2**, and the third power module **102-3**. The first surface **115** is disposed in a plane **A4**. The direction **A1** is aligned perpendicular to the plane **A4**, and a direction **A2** is perpendicular to the direction **A1**. A direction **A3** is orthogonal to directions **A1** and **A2**. As shown in FIG. 1A, the direction **A3** is depicted into the page (shown as a dot). However, since FIG. 1B is a top of the module carrier **104**,

the direction A1 on FIG. 1B is depicted into the page (as shown as a dot).

[0034] The chamber **124** may be sealed from the power modules **102** via the module carrier **104**. In other words, the sealing may not be directly on the DBC material of the power modules **102**, but rather the sealing is created using the module carrier **104**. The module carrier **104** is a single (integral) body defining a number of heat-sink slots, where the number of heat-sink slots corresponds to the number of power modules **102**. The module carrier **104** is disposed between the power modules **102** and the housing **106**. In some examples, the module carrier **104** may enable relatively easy integration with the housing **106**. In some examples, the module carrier **104** is a middle plate. In some examples, the module carrier **104** includes a plastic material. In some examples, the module carrier **104** includes a metal-based material. In some examples, the module carrier **104** includes an aluminum material. In some examples, the module carrier **104** includes a copper material. The module carrier **104** has a length that extends in the direction A2, a width that extends in the direction A3, and a thickness that extends in the direction A1. In some examples, the module carrier **104** has a rectangle shape. However, the module carrier **104** may include other types of shapes including curve and/or rounded portions.

[0035] As shown in FIG. 1B, the module carrier **104** includes a first region **114-1**, a second region **114-2**, and a third region **114-3**. The first region **114-1** is the portion of the module carrier **104** in which the first power module **102-1** is mounted. The second region **114-2** is the portion of the module carrier **104** in which the second power module **102-2** is mounted. The third region **114-3** is the portion of the module carrier **104** in which the third power module **102-3** is mounted. In some examples, if the module carrier **104** integrates three power modules **102**, the module carrier **104** may have a total of three regions, where each region has the same size. In some examples, one region may be larger than another region if a power module **102** is larger than another power module **102**.

[0036] The first region **114-1** defines a heat-sink slot **120-1**. The size (e.g., in the directions A2 and A3) of the heat-sink slot **120-1** is smaller than the size (e.g., in the directions A2 and A3) of the first region **114-1**. The second region **114-2** defines a heat-sink slot **120-2**. The size (e.g., in the directions A2 and A3) of the heat-sink slot **120-2** is smaller than the size (e.g., in the directions A2 and A3) of the second region **114-2**. The third region **114-3** defines a heat-sink slot **120-3**. The size (e.g., in the directions A2 and A3) of the heat-sink slot **120-3** is smaller than the size (e.g., in the directions A2 and A3) of the third region **114-3**.

[0037] The module carrier **104** includes a portion **126** that separates the heat-sink slot **120-1** from the heat-sink slot **120-2**. The module carrier **104** includes a portion **128** that separates the heat-sink slot **120-2** from the heat-sink slot **120-3**. The module carrier includes a perimeter area portion **118** that extends around the heat-sink slot **120-1**, the heat-sink slot **120-2**, and the heat-sink slot **120-3**, collectively. In some examples, the perimeter area portion **118** is coupled to the housing **106**. The heat-sink slot **120-1** has a size that is larger (e.g., slightly larger) than a size of the first heat sink **110-1**, and the heat-sink slot **120-1** has a size that is smaller than a size of the first power module **102-1**. The heat-sink slot **120-2** has a size that is larger (e.g., slightly larger) than a size of the second heat sink **110-2**, and the heat-sink slot **120-2** has a size that is smaller than a size of the second power module **102-2**. The heat-sink slot **120-3** has a size that is larger (e.g., slightly larger) than a size of the third heat sink **110-3**, and the heat-sink slot **120-3** has a size that is smaller than a size of the third power module **102-3**. Each of the heat-sink slots (**120-1**, **120-2**, **120-3**) extends between (and through) the first surface **115** of the module carrier **104** and the second surface **117** of the module carrier **104**.

[0038] The first region **114-1** is coupled to the first power module **102-1** such that the first heat sink **110-1** at least partially extends (or fully extends) through the heat-sink slot **120-1** and into the chamber **124** of the housing **106**. The second region **114-2** is coupled to the second power module **102-2** such that the second heat sink **110-2** at least partially extends (or fully extends) through the heat-sink slot **120-2** and into the chamber **124** of the housing **106**. The third region **114-3** is coupled

to the power module **103-3** such that the third heat sink **110-3** at least partially extends (or fully extends) through the heat-sink slot **120-4** and into the chamber **124** of the housing **106**.

[0039] As shown in FIG. **1B**, the first region **114-1** may define a positioning guide **116-1** that aligns at least one of the first power module **102-1** or the first heat sink **110-1** during assembly of the power module package **100**. In some examples, the positioning guide **116-1** is located on (or defined by) the first region **114-1** at a location proximate to the heat-sink slot **120-1**. In some examples, the positioning guide **116-1** is a feature of the heat-sink slot **120-1**. In some examples, the positioning guide **116-1** is a surface feature that interacts with a surface feature of the first power module **102-1** to cause the first power module **102-1** to be aligned with the first region **114-1** of the module carrier **104**. In some examples, the positioning guide **116-1** is a surface feature that interacts with a surface feature of the first heat sink **110-1** to cause the first heat sink **110-1** to be properly aligned within the heat-sink slot **120-1**. In some examples, the positioning guide **116-1** includes one or more protrusions. In some examples, the positioning guide **116-1** includes one or more grooves. In some examples, the positioning guide **116-1** includes one or more slots.

[0040] The second region **114-2** may define a positioning guide **116-2** that aligns at least one of the second power module **102-2** or the second heat sink **110-2**. In some examples, the positioning guide **116-2** is located on (or defined by) the second region **114-2** at a location proximate to the heat-sink slot **120-2**. In some examples, the positioning guide **116-2** is a feature of the heat-sink slot **120-2**. In some examples, the positioning guide **116-2** is a surface feature that interacts with a surface feature of the second power module **102-2** to cause the second power module **102-2** to be aligned with the second region **114-2** of the module carrier **104**. In some examples, the positioning guide **116-2** is a surface feature that interacts with a surface feature of the second heat sink **110-2** to cause the second heat sink **110-2** to be properly aligned within the heat-sink slot **120-2**. In some examples, the positioning guide **116-2** includes one or more protrusions. In some examples, the positioning guide **116-2** includes one or more grooves. In some examples, the positioning guide **116-2** includes one or more slots.

[0041] The third region **114-3** may define a positioning guide **116-3** that aligns at least one of the third power module **102-3** or the third heat sink **110-3**. In some examples, the positioning guide **116-3** is located on (or defined by) the third region **114-3** at a location proximate to the heat-sink slot **120-3**. In some examples, the positioning guide **116-3** is a feature of the heat-sink slot **120-3**. In some examples, the positioning guide **116-3** is a surface feature that interacts with a surface feature of the third power module **102-3** to cause the third power module **102-3** to be aligned with the third region **114-3** of the module carrier **104**. In some examples, the positioning guide **116-3** is a surface feature that interacts with a surface feature of the third heat sink **110-3** to cause the third heat sink **110-3** to be properly aligned with the heat-sink slot **120-3**. In some examples, the positioning guide **116-3** includes one or more protrusions. In some examples, the positioning guide **116-3** includes one or more grooves. In some examples, the positioning guide **116-3** includes one or more slots.

[0042] The housing **106** is coupled to the module carrier **104**, where the module carrier **104** is positioned between the housing **106** and the power modules **102**. In some examples, the housing **106** is coupled to the module carrier **104** via one or more fasteners. In some examples, the housing **106** is coupled to the module carrier **104** by welding (e.g., laser welding). In some examples, the housing **106** includes a plastic material. The housing **106** defines a chamber **124**, where cooling fluid is injected into the chamber **124** via an opening **125** and expelled from the chamber **124** via an opening **125**. Each of the first heat sink **110-1**, the second heat sink **110-2**, and the third heat sink **110-3** includes a portion that is disposed within the chamber **124**. In some examples, the housing **106** is an inverter housing. In some examples, the housing **106** is an inverter housing for a vehicle.

[0043] The power module package **100** includes a ring member **108** that is located (e.g., disposed) between the housing **106** and the module carrier **104**. The ring member **108** includes a portion that

contacts the housing and a portion that contacts the module carrier **104**. In some examples, the housing **106** may define a groove, and the ring member **108** is positioned in the groove between the housing **106** and the module carrier **104**. The ring member **108** extends around the first power module **102-1**, the second power module **102-2**, and the third power module **102-3**, collectively. In some examples, the ring member **108** includes an O-ring. In some examples, the power module package **100** includes a single ring member **108**.

[0044] FIGS. 2A through 2F illustrate a power module package **200** for direct cooling multiple power modules **202**. The power module package **200** may be an example of the power module package **100** of FIGS. 1A through 1C. The power module package **200** includes a plurality of power modules **202**, a plurality of heat sinks **210** coupled to the power modules **202**, a module carrier **204** coupled to the power modules **202**, a housing **206** coupled to the module carrier **204**, and a ring member **208** located (e.g., disposed) between the module carrier **204** and the housing **206**. FIG. 2A illustrates a perspective of the power module package **200**. FIG. 2B illustrates a perspective of the power module package **200** with the power modules **202** being transparent. FIG. 2C illustrates a top view perspective of the module carrier **204**. FIG. 2D illustrates a region of the module carrier **204**. FIG. 2E illustrates a cross sectional view of the power module package **200**. FIG. 2F illustrates a side view of a power module **202**.

[0045] The power modules **202** include a first power module **202-1**, a second power module **202-2**, and a third power module **202-3**. In some examples, each of the first power module **202-1**, the second power module **202-2**, and the third power module **202-3** correspond to a separate phase of a three-phase current (UVM) of an inverter. In some examples, the first power module **202-1**, the second power module **202-2**, and the third power module **202-3**, collectively, operate as a 6-pack full bridge module. In some examples, the power modules **202** includes terminals **230** and contacts **232**. In some examples, the terminals **230** extend in the direction A1, and the contacts **232** extend in the direction A2.

[0046] As shown in FIG. 2F, a power module **202** includes a first surface **211** and a second surface **213**. The distance between the first surface **211** and the second surface **213** may define a thickness of a respective power module **202** in a direction A1. Each power module **202** is connected to a separate heat sink **210**. In some examples, each power module **202** is connected to a separate heat sink **210** using an adhesive material. The first power module **202-1** is coupled to a first heat sink **210-1**, the second power module **202-2** is coupled to a second heat sink **210-2**, and the third power module **202-3** is coupled to a third heat sink **210-3**. In further detail, the first heat sink **210-1** is coupled to the second surface **213** of the first power module **202-1**, the second heat sink **210-2** is coupled to the second surface **213** of the second power module **202-2**, and the third heat sink **210-3** is coupled to the second surface **213** of the third power module **202-3**.

[0047] The first heat sink **210-1** extends from the second surface **213** of the first power module **202-1** in the direction A1. The second heat sink **210-2** extends from the second surface **213** of the second power module **202-2** in the direction A1. The third heat sink **210-3** extends from the second surface **213** of the third power module **202-3** in the direction A1. Each of the first heat sink **210-1**, the second heat sink **210-2**, and the third heat sink **210-3** includes a pin fin heat sink.

[0048] Each power module **202** is coupled to the module carrier **204** via an adhesive material **212**. The module carrier **204** includes a first surface **215** and a second surface **217**. In some examples, the second surface **217** is parallel with the first surface **215**. The distance between the first surface **215** and the second surface **217** may define the thickness of the module carrier **204** in the direction A1. The first surface **215** of the module carrier **204** is coupled to the second surface **213** of each of the first power module **202-1**, the second power module **202-2**, and the third power module **202-3**. The first surface **215** is disposed in a plane A4.

[0049] The chamber **224** may be sealed from the power modules **202** via the module carrier **204**. In other words, the sealing may not be directly on the DBC material of the power modules **102**. The module carrier **204** is a single (integral) body defining a number of heat-sink slots, where the

number of heat-sink slots corresponds to the number of power modules **202**. The module carrier **204** is disposed between the power modules **202** and the housing **206**. In some examples, the module carrier **204** is a middle plate. In some examples, the module carrier **204** includes a plastic material. In some examples, the module carrier **204** includes a metal-based material. In some examples, the module carrier **204** includes an aluminum material. In some examples, the module carrier **204** includes a copper material. The module carrier **204** has a length that extends in the direction **A2**, a width that extends in the direction **A3**, and a thickness that extends in the direction **A1**. In some examples, the module carrier **204** has a rectangle shape. However, the module carrier **204** may include other types of shapes includes curved or rounded portions.

[0050] As shown in FIG. 2C, the module carrier **104** includes a first region **214-1**, a second region **214-2**, and a third region **214-3**. The first region **214-1** defines a heat-sink slot **220-1**. The second region **214-2** defines a heat-sink slot **220-2**. The third region **214-3** defines a heat-sink slot **220-3**. The module carrier **204** includes a portion **226** that separates the heat-sink slot **220-1** from the heat-sink slot **220-2**. The module carrier **204** includes a portion **228** that separates the heat-sink slot **220-3** from the heat-sink slot **220-2**. The module carrier **204** includes a perimeter area portion **218** that extends around the heat-sink slot **220-1**, the heat-sink slot **220-2**, and the heat-sink slot **220-3**, collectively. In some examples, the module carrier **204** defines openings **236** that are used to fasten the module carrier **204** to the housing **206**. In some examples, the openings **236** are defined on the perimeter area portion **218** at the corner portions of the module carrier **204**. As shown in FIG. 2E, fasteners **205** may extend through the openings **236** and into the housing **206**. In some examples, fasteners **205** are not used, but rather the module carrier **204** is coupled to the housing **206** via other coupling mechanisms such as welding (e.g., laser welding).

[0051] The heat-sink slot **220-1** has a size that is larger than a size of the first heat sink **210-1**, and the heat-sink slot **220-1** has a size that is smaller than a size of the first power module **202-1**. The heat-sink slot **220-2** has a size that is larger than a size of the second heat sink **210-2**, and the heat-sink slot **220-2** has a size that is smaller than a size of the second power module **202-2**. The heat-sink slot **220-3** has a size that is larger than a size of the third heat sink **210-3**, and the heat-sink slot **220-3** has a size that is smaller than a size of the third power module **202-3**. Each of the heat-sink slots (**220-1**, **220-2**, **220-3**) extends between (and through) the first surface **215** of the module carrier **204** and the second surface **217** of the module carrier **204**.

[0052] The first region **214-1** is coupled to the first power module **202-1** such that the first heat sink **210-1** extends through the heat-sink slot **220-1** and into the chamber **224** of the housing **206**. The second region **214-2** is coupled to the second power module **202-2** such that the second heat sink **210-2** extends through the heat-sink slot **220-2** and into the chamber **224** of the housing **206**. The third region **214-3** is coupled to the power module **203-3** such that the third heat sink **210-3** extends through the heat-sink slot **220-4** and into the chamber **224** of the housing **206**.

[0053] As shown in FIG. 2D, in some examples, the first region **214-1** may define module positioning guides **240**. The module positioning guides **240** may be an example of the positioning guide **116-1** of FIG. 1B. The module positioning guides **240** are surface features configured to interact with surface features on the first power module **202-1** to properly align the first power module **202-1** with the first region **214-1**. Although four module positioning guides **240** are depicted in FIG. 2D, the first region **214-1** may define any number of module positioning guides **240**. In some examples, the module positioning guides **240** include protrusions that extend from the second surface **217** of the module carrier **204**, where the protrusions interact with slots defined by the first power module **202-1**. In some examples, the module positioning guides **240** include slots that extend into the second surface **217** of the module carrier **204**, where the slots interact with protrusions defined by the first power module **202-1**.

[0054] In some examples, the first region **214-1** may include heat-sink positioning guides **242**. The heat-sink positioning guides **242** may be an example of the positioning guide **116-1** of FIG. 1B. In some examples, the first region **214-1** includes the module positioning guides **240** but not the heat-

sink positioning guides **242**. In some examples, the first region **214-1** includes the heat-sink positioning guides **242** but not the module positioning guides **242**. In some examples, the first region **214-1** includes the module positioning guides **240** and the heat-sink positioning guides **242**. [0055] The heat-sink positioning guides **242** are surface features configured to interact with surface features on the first heat sink **210-1** to properly align the first heat sink **210-1** within the heat-sink slot **220-1**. Although six heat-sink positioning guides **242** are depicted in FIG. 2D, the first region **214-1** may define any number of heat-sink positioning guides **242**. In some examples, the heat-sink positioning guides **242** include slots that extend from the first heat-sink slot **220-1**. The second region **214-2** and the third region **214-3** may include the same features as discussed with reference to the first region **214-1**. In some examples, the first region **214-1** includes one or more adhesive stop features **244**. In some examples, the adhesive stop features **244** are slots that extend into the first region **214-1** (e.g., having a certain depth). The adhesive stop features **244** may accept adhesive overflow during the assembly process.

[0056] The housing **206** is coupled to the module carrier **204**, where the module carrier **204** is positioned between the housing **206** and the power modules **202**. In some examples, the housing **206** is coupled to the module carrier **204** via one or more fasteners **205**. In some examples, the housing **206** is coupled to the module carrier **204** by welding (e.g., laser welding). In some examples, the housing **206** includes a plastic material. The housing **206** defines a chamber **224**, where cooling fluid is injected into the chamber **224** via an opening **225** and expelled from the chamber **224** via an opening **225**. Each of the first heat sink **210-1**, the second heat sink **210-2**, and the third heat sink **210-3** includes a portion that is disposed within the chamber **224**. In some examples, the housing **206** is an inverter housing. In some examples, the housing **206** is an inverter housing for a vehicle.

[0057] The power module package **200** includes a ring member **208** that is located (e.g., disposed) between the housing **206** and the module carrier **204**. The ring member **208** includes a portion that contacts the housing and a portion that contacts the module carrier **204**. In some examples, the housing **206** may define a groove **271**, and the ring member **208** is positioned in the groove **271** between the housing **206** and the module carrier **204**. The ring member **208** extends around the first power module **202-1**, the second power module **202-2**, and the third power module **202-3**, collectively. In some examples, the ring member **208** includes an O-ring. In some examples, the power module package **200** includes a single ring member **208**.

[0058] FIG. 3 illustrates an example of a heat sink **310**. The heat sink **310** may be any of the heat sinks discussed with reference to the previous figures. The heat sink **310** is a pin fin heat sink. For example, the heat sink **310** includes a base plate **323** and a plurality of pin fins **321** that extend from the base plate **323**. In some examples, as shown in FIG. 3, the pin fins **321** have the same length. FIG. 4 illustrates an example of a heat sink **410**. The heat sink **410** may be any of the heat sinks discussed with reference to the previous figures. The heat sink **410** is a pin fin heat sink. For example, the heat sink **410** includes a base plate **423** and a plurality of pin fins **421** that extend from the base plate **423**. In some examples, as shown in FIG. 4, the pin fins **421** have different lengths.

[0059] FIGS. 5A through 5F illustrate a power module package **500** that is assembled according to a first assembly process. The power module package **500** may be an example of any of the power module packages discussed with reference to the previous figures. FIG. 5A illustrates a power module **502** with an attached heat sink **510**, which may be an example of a first power module **502-1**, a second power module **502-2**, and a third power module **502-3**. Also, the power module **502** includes terminals **530** and contacts **532**. FIG. 5B illustrates a module carrier **504** defining a first region **514-1**, a second region **514-2**, and a third region **514-3**. The first region **514-1** defines a first heat-sink slot **520-1**. The second region **514-2** defines a second heat-sink slot **520-2**. The third region **514-3** defines a third heat-sink slot **520-3**. Also, the module carrier **504** may include a plurality of openings **536**. FIG. 5C illustrates a perspective of the first region **514-1** that defines

module positioning guides **540**, heat-sink positioning guides **542**, and/or adhesive stop features **544**. FIG. 5D illustrates an adhesive material **512** applied to the first region **514-1**, where the adhesive stop features **544** accept adhesive overflow from the application of the adhesive material **512** to the first region **514-1**. FIG. 5E illustrates the power module **502** coupled to the first region **514-1**. FIG. 5F illustrates a first power module **502-1**, a second power module **502-2**, and a third power module **502-3** coupled to the module carrier **504** after the adhesive material **512** is cured. [0060] Before the first assembly process starts, the heat sinks **510** are already coupled to the power modules **502**. With respect to the first region **514-1**, an adhesive material **512** is applied to the first region **514-1**. Then, the heat sink **510** that is coupled to the first power module **502-1** is inserted into the first heat-sink slot **520-1**, where the heat-sink positioning guides **542** guide the heat sink **510** through the heat-sink slot **520-1** such that the first power module **502-1** contacts the adhesive material **512**. The same process happens with respect to the second region **514-2** and the third region **514-3**. For example, the adhesive material **512** is applied to the first region **514-1**, the second region **514-2**, and the third region **514-3**. Then, the first power module **502-1**, the second power module **502-2**, and the third power module **502-3** are positioned on the module carrier **504** in the same manner as discussed with reference to the first region **514-1**. Then, the module carrier **504** with the first power module **502-1**, the second power module **502-2**, and the third power module **502-3** may be cured (e.g., oven cured) to cure the adhesive material **512**.

[0061] FIG. 6 depicts a flowchart **600** depicting example assembly operations for assembling a power module package according to an aspect. Although the flowchart **600** is described with reference to the power module package **100**, the flowchart may be applicable to any of the power module packages discussed herein. Although the flowchart **600** of FIG. 6 illustrates operations in sequential order, it will be appreciated that this is merely an example, and that additional or alternative operations may be included. Further, operations of FIG. 6 and related operations may be executed in a different order than that shown, or in a parallel or overlapping fashion.

[0062] Operation **602** includes providing a module carrier **104**, where the module carrier **104** includes a first region **114-1** defining a first heat-sink slot **120-1** and a second region **114-2** defining a second heat-sink slot **120-2**. Operation **604** includes applying an adhesive material **112** to the first region **114-1** and the second region **114-2**. Operation **606** includes inserting a first heat sink **110-1** coupled to a first power module **102-1** through the heat-sink slot **120-1** until the first power module **102-1** contacts the adhesive material **112**. Operation **608** includes inserting a second heat sink **110-2** coupled to a second power module **102-2** through the heat-sink slot **120-2** until the second power module **102-2** contacts the adhesive material **112**.

[0063] FIGS. 7A through 7D illustrate a power module package that is assembled according to a second assembly process. The power module package of FIGS. 7A through 7D may be an example of any of the power module packages discussed with reference to the previous figures. FIG. 7A illustrates a module carrier **704** defining a first region **714-1**, a second region **714-2**, and a third region **714-3**. The first region **714-1** defines a first heat-sink slot **720-1**. The second region **714-2** defines a second heat-sink slot **720-2**. The third region **714-3** defines a third heat-sink slot **720-3**. In contrast to FIGS. 5A through 5F, the heat sinks (e.g., a first heat sink **710-1**, a second heat sink **710-2**, a third heat sink **710-3**) are not coupled to the power modules at the start of the second assembly process. Rather, the first heat sink **710-1** is inserted into the first heat-sink slot **720-1**, the second heat sink **710-2** is inserted into the second heat-sink slot **720-2**, and the third heat sink **710-3** is inserted into the third heat-sink slot **720-3**. FIG. 7B illustrates a perspective of the first region **714-1** that defines module positioning guides **740**, heat-sink positioning guides **742**, and/or adhesive stop features **744**. FIG. 7C illustrates an adhesive material **512** applied to the first region **514-1** and the first heat sink **710-1**. In some examples, separate (small) portions **760** are applied to the first heat sink **710-1**. In some examples, a single large portion **760** is applied to the first heat sink **710-1**. FIG. 7D illustrates a power module **702** coupled to the first region **714-1** and the first heat sink **710-1** via the adhesive material **712**.

[0064] Before the second assembly process starts, the first heat sink **710-1**, the second heat sink **710-2**, the third heat sink **710-3** are not coupled to the power modules. Rather, the first heat sink **710-1**, the second heat sink **710-2**, the third heat sink **710-3** are inserted into the first heat-sink slot **720-1**, the second heat-sink slot **720-2**, and the third heat-sink slot **720-3**, respectively. Then, the adhesive material **712** is applied to each of the first region **714-1**, the second region **714-2**, and the third region **714-3**, the first heat sink **710-1**, the second heat sink **710-2**, and the third heat sink **710-3**. Then, a first power module is attached to the first region **714-1** and the first heat sink **710-1**, a second power module is attached to the second region **714-2** and the second heat sink **710-2**, and a third power module is attached to the third region **714-3** and the third heat sink **710-3**. Then, the module carrier **704** with the first power module, the second power module, and the third power module may be cured (e.g., oven cured) to cure the adhesive material **712**.

[0065] FIG. **8** depicts a flowchart **800** depicting example assembly operations for assembling a power module package according to an aspect. Although the flowchart **800** is described with reference to the power module package **100**, the flowchart may be applicable to any of the power module packages discussed herein. Although the flowchart **800** of FIG. **8** illustrates operations in sequential order, it will be appreciated that this is merely an example, and that additional or alternative operations may be included. Further, operations of FIG. **8** and related operations may be executed in a different order than that shown, or in a parallel or overlapping fashion.

[0066] Operation **802** includes providing a module carrier **104**, where the module carrier **104** includes a first region **114-1** defining a first heat-sink slot **120-1** and a second region **114-2** defining a second heat-sink slot **120-2**. Operation **804** includes inserting a first heat sink **110-1** into the first heat-sink slot **120-1**. Operation **806** includes inserting a second heat sink **110-2** into the second heat-sink slot **120-2**. Operation **808** includes applying an adhesive material **112** to the first region **114-1**, the second region **114-2**, the first heat sink **110-1**, and the second heat sink **110-2**. Operation **810** includes contacting a first power module **102-1** to the adhesive material **112** on the first region **114-1** and the first heat sink **110-1**. Operation **812** includes contacting a second power module **102-2** to the adhesive material **112** on the second region **114-2** and the second heat sink **110-2**.

[0067] It will be understood that, in the foregoing description, when an element is referred to as being connected to, electrically connected to, coupled to, or electrically coupled to another element, it may be directly connected or coupled to the other element, or one or more intervening elements may be present. In contrast, when an element is referred to as being directly connected to or directly coupled to another element, there are no intervening elements. Although the terms directly connected to, or directly coupled to may not be used throughout the detailed description, elements that are shown as being directly connected or directly coupled can be referred to as such. The claims of the application, if any, may be amended to recite exemplary relationships described in the specification or shown in the figures. Implementations of the various techniques described herein may be implemented in (e.g., included in) digital electronic circuitry, or in computer hardware, firmware, software, or in combinations of them. Portions of methods also may be performed by, and an apparatus may be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

[0068] Some implementations may be implemented using various semiconductor processing and/or packaging techniques. Some implementations may be implemented using various types of semiconductor processing techniques associated with semiconductor substrates including, but not limited to, for example, Silicon (Si), Gallium Arsenide (GaAs), Gallium Nitride (GaN), Silicon Carbide (SiC) and/or so forth.

[0069] While certain features of the described implementations have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the scope of the embodiments. It should be understood that they have been presented by way of example only, not limitation, and various changes in form

and details may be made. Any portion of the apparatus and/or methods described herein may be combined in any combination, except mutually exclusive combinations. The embodiments described herein can include various combinations and/or sub-combinations of the functions, components and/or features of the different embodiments described.

Claims

1. A power module package comprising: a plurality of power modules; a plurality of heat sinks coupled to the plurality of power modules; a module carrier coupled to the plurality of power modules, the module carrier including a plurality of regions, a region of the plurality of regions defining a heat-sink slot, a heat sink of the plurality of heat sinks extending at least partially through the heat-sink slot; and a housing coupled to the module carrier.
2. The power module package of claim 1, wherein the heat sink includes a plurality of metal pin fins.
3. The power module package of claim 1, wherein the region includes at least one positioning guide configured to align a power module of the plurality of power modules.
4. The power module package of claim 1, wherein the module carrier includes a polymer material.
5. The power module package of claim 1, wherein the region is a first region, wherein the plurality of power modules include a first power module, a second power module, and a third power module, wherein the plurality of regions include a second region and a third region, wherein the first power module, the second power module, and the third power module are coupled to the first region, the second region, and the third region, respectively.
6. The power module package of claim 5, wherein each of the first power module, the second power module, and the third power module correspond to a separate phase of a three phase current of an inverter.
7. The power module package of claim 1, wherein the housing includes a chamber configured to receive cooling fluid.
8. The power module package of claim 1, wherein the module carrier is disposed between the housing and the plurality of power modules.
9. The power module package of claim 1, wherein the module carrier is coupled to the housing via one or more fasteners.
10. The power module package of claim 1, further comprising: a ring member located between the module carrier and the housing.
11. The power module package of claim 10, wherein the ring member includes a closed loop that extends around the plurality of power modules, collectively.
12. A power module package comprising: a plurality of power modules; a plurality of heat sinks coupled to the plurality of power modules; a module carrier coupled to the plurality of power modules, the module carrier including a plurality of regions, a region of the plurality of regions defining a heat-sink slot, a heat sink of the plurality of heat sinks extending at least partially through the heat-sink slot; and a housing coupled to the module carrier, the housing including a chamber configured to receive cooling fluid, the heat sink having a portion that extends into the chamber.
13. The power module package of claim 12, wherein the module carrier is located between the housing and the plurality of power modules.
14. The power module package of claim 12, further comprising: a ring member located between the module carrier and a surface of the housing, wherein the ring member includes a closed loop that extends around the plurality of power modules, collectively.
15. The power module package of claim 12, wherein the region includes one or more module positioning guides, wherein the region includes one or more heat-sink positioning guides.
16. The power module package of claim 12, wherein the module carrier includes a metal-based

material.

17. A power module package comprising: a plurality of power modules; a plurality of heat sinks coupled to the plurality of power modules; a module carrier coupled to the plurality of power modules, the module carrier including a plurality of heat-sink slots, a heat sink of the plurality of heat sinks extending at least partially through a heat-sink slot of the plurality of heat-sink slots; a housing coupled to the module carrier; and a ring member located between the module carrier and the housing.

18. The power module package of claim 17, wherein the housing includes a chamber configured to receive cooling fluid, the heat sink having a portion that extends into the chamber.

19. The power module package of claim 17, wherein the module carrier includes one or more module positioning guides, wherein the module carrier includes one or more heat-sink positioning guides.

20. The power module package of claim 17, wherein the plurality of power modules includes a first power module, a second power module, and a third power module.
